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```
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## Experiment 6 : Low Pass Filtering (Smoothing) in Spatial Domain

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```
clc; close all; clear;
```

### Simple Averaging

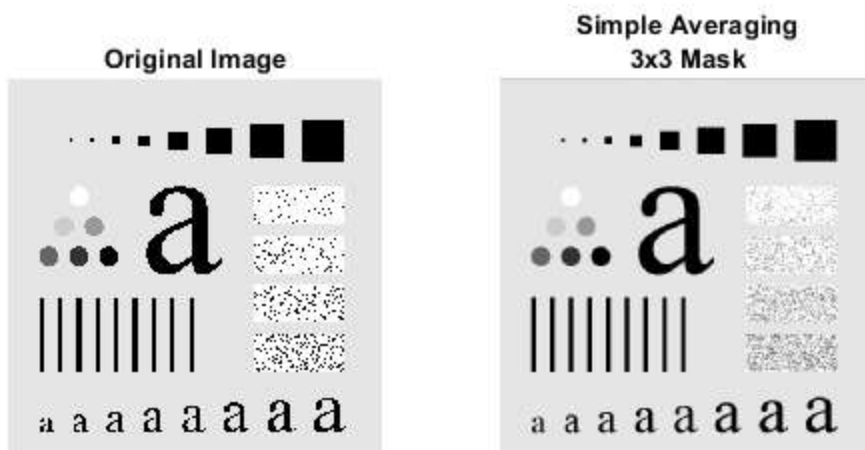
---

```
I = im2double(imread('Images3/a_blurring_ch3.tif'));

m3x3 = ones(3, 3) / 9;

g = imfilter(I, m3x3, 'same');

figure,
subplot(1, 2, 1), imshow(I), title("Original Image");
subplot(1, 2, 2), imshow(g, []), title({"Simple Averaging", "3x3 Mask"});
```

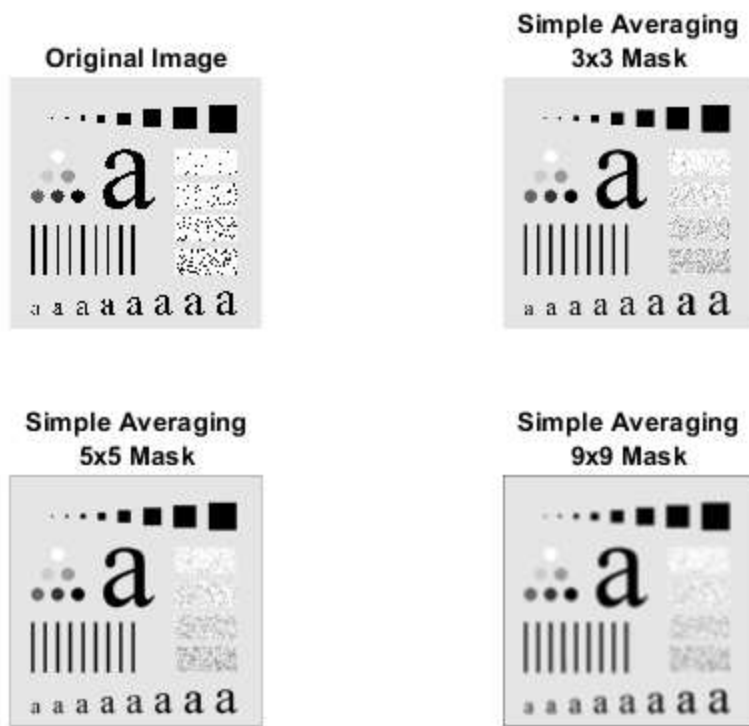


### Simple Averaging using 3x3, 5x5 and 9x9 masks.

```
m5x5 = ones(5, 5) / 25;
m9x9 = ones(9, 9) / 81;

g1 = imfilter(I, m5x5, 'same');
g2 = imfilter(I, m9x9, 'same');

figure,
subplot(2, 2, 1), imshow(I), title("Original Image");
subplot(2, 2, 2), imshow(g, []), title({"Simple Averaging", "3x3 Mask"});
subplot(2, 2, 3), imshow(g1, []), title({"Simple Averaging", "5x5 Mask"});
subplot(2, 2, 4), imshow(g2, []), title({"Simple Averaging", "9x9 Mask"});
```

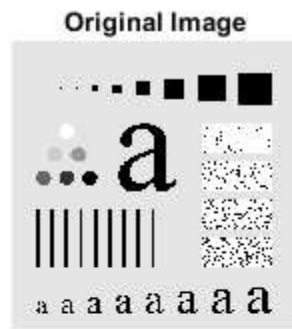


## Weighted Averaging

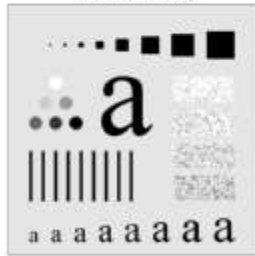
```
mw5x5 = [1 1 2 1 1; 1 2 4 2 1; 2 4 8 4 2; 1 2 4 2 1; 1 1 2 1 1];
sum = sum(mw5x5(:));
mw5x5 = mw5x5 / sum;

gw1 = imfilter(I, mw5x5, 'same');

figure,
subplot(2, 2, [1 2]), imshow(I), title("Original Image");
subplot(2, 2, 3), imshow(g1, []), title({"Simple Averaging", "5x5 Mask"});
subplot(2, 2, 4), imshow(gw1, []), title({"Weighted Averaging", "5x5 Mask"});
```



Simple Averaging  
5x5 Mask



Weighted Averaging  
5x5 Mask



## Conclusion

The input images were effectively smoothened using both Simple and Weighted Averaging filters. As the filter size increased, the blurring effect also became more pronounced. Weighted Averaging demonstrated superior enhancement compared to Simple Averaging, as it better preserves important details in the images.